application taking 0.15 hours for a total of 0.3 hours per competing consolidator. On an ongoing basis, each competing consolidator will add one individual to access EFFS for amendments, adding 0.15 hours per competing consolidator. To make a submission into EFFS, the competing consolidator must download a proprietary viewer.

Because EFFS is not available to the public, when the Commission makes an effective Form CC available to the public, the Commission will transform the data into an unstructured format, meaning that it is not machine-readable. Market participants that seek to use the Form CC data to evaluate and compare competing consolidators will bear the costs of locating, comparing, and evaluating the information on the Commission's website and take steps to put the information "side by side" for comparison purposes.

The Commission believes that the public disclosure of performance metrics and additional information will introduce transparency to the operations of competing consolidators. These metrics should allow subscribers and potential subscribers to better evaluate the performance and current and future capabilities of a competing consolidator. Market participants, based on their individual needs, will be able to review competing consolidators' performance statistics and choose ones that will best serve their trading needs. While the requirements to post the monthly performance metrics and operational information on websites will introduce transparency, they will not completely eliminate costs incurred when market participants want to compare competing consolidators because collecting the information will involve market participants expending some resources to go to each competing consolidator's website.

Competing consolidators will also incur implementation and ongoing compliance costs in order to setup and maintain systems required to calculate and produce the information for the

performance metrics as well as other information the competing consolidator will be required to post to its website.

Each month, competing consolidators will be required to post the monthly performance metrics and operational information on their own websites. Excluding the cost of preparing the information, the Commission estimates an average competing consolidator will incur a one-time cost of $2,651 (6 hours (for website development) × $308.50 per hour (blended rate for a senior systems analyst ($285) and senior programmer ($332)) + $800 for an external website developer to develop the web page = $2,651) for posting the required information to a website, and will incur an ongoing annual cost of up to $3,702 (1 hour (for website updates) × $308.50 per hour (blended rate for a senior systems analyst ($285) and senior programmer ($332)) x 12 monthly postings = $3,702) to update the relevant web page each month. Because the monthly performance metrics and operational information may be posted in any format the competing consolidator finds most convenient, market participants that seek to use the data to evaluate and compare competing consolidators will bear the costs of locating, comparing, and evaluating the information on each competing consolidator's website.

The Commission believes that the operational information that competing consolidators will be required to publicly disclose on their websites will create a mechanism for market participants to hold competing consolidators accountable for any systems issues they may experience. One strong accountability mechanism market participants have is their purchasing power. The disclosure requirements will alert market participants to any system breaches or any data quality or systems issues a competing consolidator experiences. Market participants could hold competing consolidators accountable by abandoning competing consolidators that repeatedly experience system issues and gravitating toward competing consolidators that

demonstrate more reliable systems through their disclosures. This demand shift could cause competing consolidators with less reliable systems to exit the market.

In addition to the requirements of Rule 614(d)(9) and Regulation SCI promoting competing consolidators to develop resilient systems,[2379] the requirement that competing consolidators publicly disclose information on systems issues as well as performance metrics regarding system availability could also encourage competing consolidators to make investments that will ensure the resiliency of their systems. These disclosures will help market participants determine which competing consolidators have more reliable systems. Competing consolidators who display more reliable systems with greater system availability will attract more subscribers. This should incentivize competing consolidators to invest in better backup systems or other technology that will improve the resiliency of their systems and increase their system uptime.

The Commission believes that information from the disclosures in Form CC and the performance metrics and operational information competing consolidators will provide on their websites will promote effective regulatory oversight of competing consolidators and increased investor protection by providing the Commission and relevant SROs with information about competing consolidators. With this information, the Commission and the SROs could identify competing consolidators that are not properly complying with the final amendments or parts of them. The Commission and SROs, then, could utilize this information to help prioritize examinations and possibly help identify potential issues.

The Commission believes that the public disclosure of the information in Form CC on the Commission's website and the public disclosure of performance metrics and operational information on competing consolidators' websites could also increase competition between

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[2379] See supra Section V.C.2(e)(i).

competing consolidators and also expose some competing consolidators to certain competitive effects.[2380] If the public disclosures show that certain competing consolidators have higher fees or poorer performance, it may result in those competing consolidators losing subscribers and earning lower revenues. Similarly, competing consolidators who display lower prices or superior system performance may be able to attract more subscribers and earn more revenue. The public disclosure of the fee and performance information on the Commission and competing consolidator websites will facilitate competing consolidator comparison and will also promote competition. Greater competition between competing consolidators could in turn incentivize competing consolidators to innovate—particularly in terms of their technology—so that they can attract more subscribers.[2381]

As discussed above, Rule 614(d)(9)(iii)(C) will require a competing consolidator that is not an SCI competing consolidator to notify the Commission and provide the Commission with updates on Form CC if it experiences a systems disruption or intrusion.[2382] The Commission believes that this information will help ensure more effective Commission oversight of competing consolidators by helping make the Commission aware of potential areas of weakness in the competing consolidator's systems that may pose a risk to the entity or the market as a whole.[2383]

2380 A commenter agreed the public disclosure of Form CC and monthly performance metrics would enhance competition between competing consolidators. See Clearpool Letter at 9.

2381 See infra Section V.D.2 discussing the potential effects of the proposal on competition.

2382 See supra Section III.F.

2383 See supra Section V.C.2(e)(i).

(b) Commission Review and Process for Declaring Initial Form CC Ineffective

The Commission believes that the process of reviewing an initial Form CC will allow the Commission to evaluate, among other things, the completeness and comprehensibility of a competing consolidators' disclosures and, if necessary, declare the Form CC ineffective. To be a consolidated market data provider, a competing consolidator is required to have a Form CC that has become effective pursuant to Rule 614(a)(1)(v). Thus, for competing consolidators that submit low quality and potentially inaccurate data, the Commission's review and declaration of their Form CC ineffective could start an iterative cycle of increasingly better information provision, until the competing consolidator can have an effective Form CC. The Commission believes that this public disclosure and review process will improve the quality of information the Commission receives from competing consolidators, which will allow the Commission to better protect investors from potentially incomprehensible or incomplete disclosures that will misinform market participants about the operations of the competing consolidator. Additionally, an entity cannot operate as a competing consolidator without an effective Form CC. The Commission's review will be designed to ensure that the competing consolidators serving the investors will be the ones that meet the Commission's qualification requirements.

The Commission believes that the filing requirements of Form CC and the Commission review period could impose costs on competing consolidators. The Commission believes that declaring a Form CC ineffective could impose costs on a competing consolidator—such as delaying the start of operations while the competing consolidator refiles its Form CC—and could impose costs on individual market participants and the overall market for competing consolidators resulting from a potential reduction in competition. However, competing consolidators and market participants will not incur these costs unless the competing

consolidator filed a deficient Form CC. Therefore, the Commission believes that a competing consolidator will be incentivized to file Form CC disclosures that are complete and comprehensive to avoid bearing the costs of refiling a Form CC filing or of having its Form CC declared ineffective.

The Commission recognizes that the registration process will create uncertainty about whether the form will be declared ineffective. This uncertainty may create a disincentive for entities to become competing consolidators, which could potentially reduce competition in the competing consolidator market.[2384]

4. Economic Effects from the Interaction of Changes to Core Data and the Decentralized Consolidation Model

The Commission believes that the final amendments would have a number of economic effects that are only possible as a result of a combination of the expanded content of core data and latency reductions due to the introduction of the decentralized consolidation model.[2385] Specifically, the Commission believes that the combination of these factors would affect proprietary data feed business; market participants who choose to engage in market making, smart order routing, and other latency sensitive trading businesses; the Consolidated Audit Trail; and data vendor business.

(a) Economic Effects on the Proprietary Data Feed Business

The Commission believes that the expanded content of core data and latency reduction due to the introduction of the decentralized consolidation model could make consolidated market data a reasonable alternative to exchange proprietary data feeds for some market participants.

2384 See infra Section V.D.2 (discussing the potential effects of the proposal on competition).

2385 See supra Section V.C.2(c) discussing the effect of the decentralized consolidation model on consolidated market data latency.

This would have the effect of providing these market participants with a potentially lower cost option (relative to proprietary feeds) for low-latency, high-content market data. The lower cost of either self-aggregating consolidated market data or obtaining a competing consolidator's data feed will come at the expense of losing the full set of data currently available via proprietary feeds, because the consolidated market data definition does not include all data elements currently available via proprietary data feeds.[2386] Nevertheless, some market participants may find that the expanded content of core data makes the trade-off worth it and may choose to drop their proprietary feed subscriptions in favor of the consolidated market data.

This effect will represent a transfer from exchanges who sell proprietary data feeds to the market participants who would save money by either self-aggregating consolidated market data or subscribing to a competing consolidator's data feed. In the latter case, a portion of the benefit is also transferred to the competing consolidator in the form of additional business. The Commission believes that a transfer from the exchanges to market participants may help market participants enhance their product and service offerings to their customers. Additional business and revenues for competing consolidators may enhance competing consolidators' efforts to offer higher quality products and a wider range of product offerings.[2387]

It is possible that changes to the pricing and customer base of core and proprietary data feeds may not have a uniform impact across all exchanges. Some exchanges currently have more proprietary feed revenue than others, and some exchanges may currently rely more on

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2386 Commenters agreed that switching to new consolidated market data would come with this expense of losing some data compared to the proprietary data feeds. One commenter stated that it would be unable to remain competitive even after the final amendments are in place without continuing to purchase proprietary data feeds. See Virtu Letter at 2. See also Clearpool Letter at 3, supporting the idea that there may be broker-dealers who will still need proprietary feeds.

2387 See supra Section V.C.2(c).

revenue from SIP data fees than other exchanges. To the extent that an exchange receives a large share of revenue from its proprietary feed business, the impact of these potential reductions in proprietary feed subscriptions could be large for that exchange. To the extent that an exchange receives only a small portion of its revenue from proprietary feed subscriptions, the impact of these potential reductions in subscriptions could be small for that exchange.

The Commission also notes that the exchanges' revenues from connectivity services may increase or decrease, depending on any new data connectivity fees that the exchanges may propose for data content use cases. The connectivity fees for consolidated market data must be fair and reasonable and not unreasonably discriminatory.[2388] If these new connectivity fees are higher than current fees, there is a possibility that the exchanges' overall revenue from connectivity services would increase. It is also possible that exchanges could lose revenue from existing customers reducing the number of ports or the amount of bandwidth they purchase as they switch to competing consolidators for some use cases. The overall effect on the exchanges' connectivity revenues is uncertain, and the impact on connectivity revenues could differ across different exchanges.

The Commission believes that these competitive pressures on the exchange proprietary feed and connectivity business could also have the effect of causing the exchanges to lower the fees they charge for these services in an effort to stay competitive with the consolidated market data. This effect represents a transfer from the exchanges to the customers of these services. To the extent that existing customers of these services invest the money saved from lower fees in new products (such as expanding brokerage services) this effect will also have benefit of encouraging the creation of new products and services. To the extent that the lower fees for

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[2388] See supra note 1134.

these services enable new market participants to subscribe to these feeds and offer the services that these feeds are required for (such as high quality execution brokerage services), this effect will also represent a benefit in the form of new competition in the broker-dealer business.

One commenter stated that the final amendments would have the effect of increasing proprietary data fees, because "demand is inelastic."[2389] The Commission acknowledges that if some market participants no longer purchase proprietary data feeds after the rule is implemented, those who continue to purchase proprietary data feeds are likely to value those feeds more than the ones who no longer make these purchases. This means that the exchanges could infer that their remaining proprietary customers might actually be willing to pay more for the data then their old customer base, and consequently attempt to increase proprietary fees. However, the Commission believes that the need to remain competitive against new consolidated market data could overwhelm the effect of knowing that remaining customers might be willing to pay more. If this is the case, then the exchanges will instead lower their prices for proprietary data.

If exchanges increase proprietary fees as a result of these potential insights into the demand elasticity of the remaining customer base after the rules are implemented, it will result in a transfer from those market participants who continue to purchase proprietary data to the exchanges, while any market participants who stop purchasing proprietary data as a result of the fee increases will represent an economic cost. The Commission is uncertain as to whether fees will increase or decrease for proprietary data.

[2389] See Data Boiler Letter I at 2. For further support that proprietary fees could increase, see Clearpool Letter at 3.

The Commission believes, however, that if a small latency differential between competing consolidator feeds and the proprietary data feeds remains, then the above effects are likely to be small, owing to the nature of high speed competition.[2390] However, this limitation would only be for the case where current subscribers to proprietary data feeds switch to using a competing consolidator feed. In the case of those proprietary feed subscribers who become self-aggregators, the Commission believes that it is unlikely that this would result in a latency differential compared to receiving proprietary data.[2391] It is also possible that the data that would remain exclusive to proprietary feeds would also reduce the incentives for market participants to switch to using consolidated market data only, further reducing the size of the above effects.

In the event that proprietary data feed subscribers are willing to switch to receiving new consolidated market data products and a latency differential remains between these feeds and feeds provided by competing consolidators, the effects discussed in this section would apply only to those market participants who become self-aggregators. The Commission believes that the set of current subscribers of proprietary feeds willing to become self-aggregators may be smaller than the set of current subscribers willing to switch to using a competing consolidator, as it is possible that subscribing to a competing consolidator would be more convenient or less costly. To the extent this is the case, the size of the effects described in this section will be reduced.

2390 See supra Section V.B.2(b).

2391 More generally, the final rule could enable some reduction in the latency differential between current market participants to the extent that such market participants would be willing to make the necessary technology and personnel investments to take advantage of the latency reductions provided by the decentralized consolidation model. Thus, while some differences in latency may remain, the barriers to entry for market participants to compete in the latency sensitive businesses at various levels of sophistication and competitiveness would be reduced. See also Sections V.B.2(f) and V.C.4(b) for further discussion of this point.